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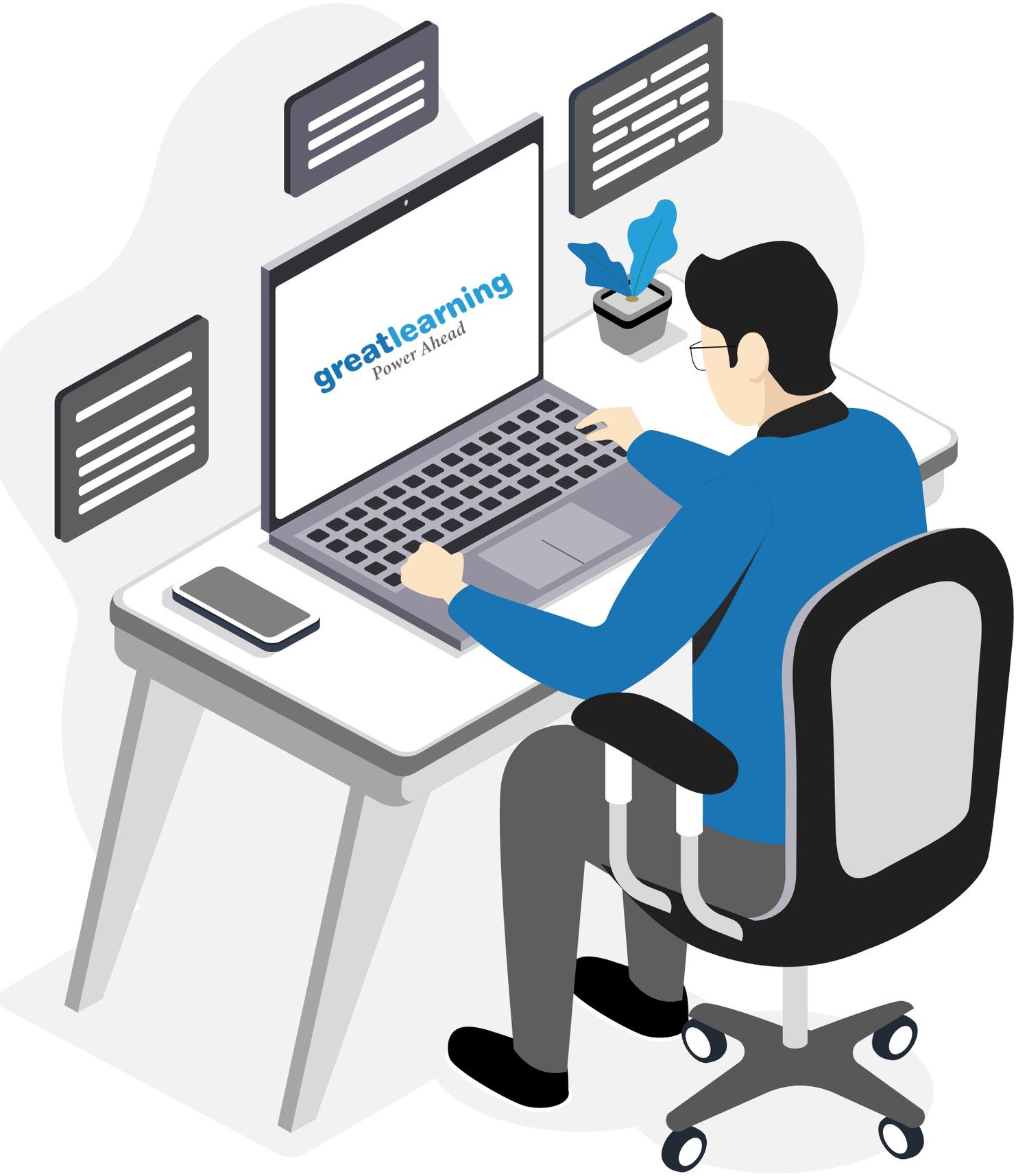
Power Ahead



AIML | MODULE PROJECT

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1 AIML module projects are designed to have a detailed hands on to integrate theoretical knowledge with actual practical implementations.

2 AIML module projects are designed to enable you as a learner to work on realtime industry scenarios, problems and datasets.

3 AIML module projects are designed to enable you simulating the designed solution using AIML techniques onto python technology platform.

4 AIML module projects are designed to be scored using a predefined rubric based system.

5 AIML module projects are designed to enhance your learning above and beyond. Hence, it might require you to experiment, research, self learn and implement.

AIML | MODULE PROJECT

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NEURAL NETWORKS

PART I

AIML module project part I consists of industry based problem which can be solved using neural networks as a regressor.

PART III

AIML module project Part III consists of implementing a clickable GUI which can automate Part I & II of this project.

PART II

AIML module project part II consists of industry based problem which can be solved using neural networks as a classifier.

TOTAL SCORE 30

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PART ONE | PROJECT BASED | TOTAL SCORE 10

- **DOMAIN:** Electronics and Telecommunication
- **CONTEXT:** A communications equipment manufacturing company has a product which is responsible for emitting informative signals. Company wants to build a machine learning model which can help the company to predict the equipment's signal quality using various parameters.
- **DATA DESCRIPTION:** The data set contains information on various signal tests performed:
    1. Parameters: Various measurable signal parameters.
    2. Signal_Quality: Final signal strength or quality
- **PROJECT OBJECTIVE:** The need is to build a regressor which can use these parameters to determine the signal strength or quality [as number].

Steps and tasks: **[Total Score: 10 points]**

1. Import data.
2. Data analysis & visualisation
    - Perform relevant and detailed statistical analysis on the data.
    - Perform relevant and detailed uni, bi and multi variate analysis.

    Hint: Use your best analytical approach. Even you can mix match columns to create new columns which can be used for better analysis. Create your own features if required. Be highly experimental and analytical here to find relevant hidden patterns.
3. Design, train, tune and test a neural network regressor.

    Hint: Use best approach to refine and tune the data or the model. Be highly experimental here.
4. Pickle the model for future use.

PART TWO | PROJECT BASED | TOTAL SCORE 10

- **DOMAIN:** Electronics and Telecommunication
- **CONTEXT:** A communications equipment manufacturing company has a product which is responsible for emitting informative signals. Company wants to build a deep learning model which can help the company to predict the equipment's signal quality using various parameters.
- **DATA DESCRIPTION:** The data set contains information on various signal tests performed:
    1. Parameters: Various measurable signal parameters.
    2. Signal_Quality: Final signal strength or quality
- **PROJECT OBJECTIVE:** The need is to build a classifier which can use these parameters to determine the signal strength or quality [as number].

Steps and tasks: **[Total Score: 10 points]**

1. Import data
2. Data analysis & visualisation
    - Perform relevant and detailed statistical analysis on the data.
    - Perform relevant and detailed uni, bi and multi variate analysis.

    Hint: Use your best analytical approach. Even you can mix match columns to create new columns which can be used for better analysis. Create your own features if required. Be highly experimental and analytical here to find hidden patterns.
3. Design, train, tune and test a neural network classifier..

    Hint: Use best approach to refine and tune the data or the model. Be highly experimental here.
4. Pickle the model for future use.

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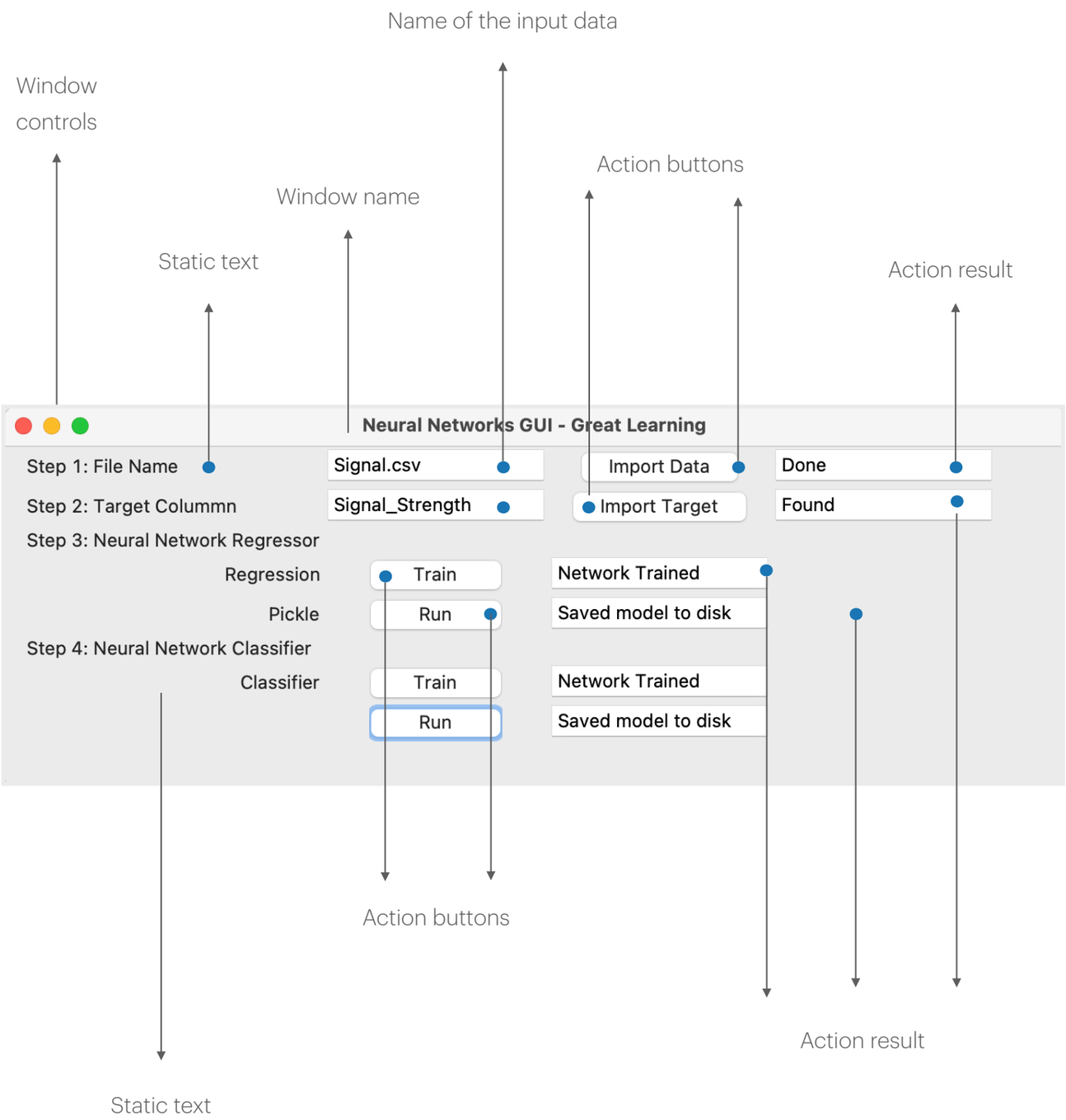
PART THREE | PROJECT BASED | TOTAL SCORE 10

- **CONTEXT:** Develop a clickable GUI [desk application or web service application] which can automate Part I & II of this project.



The above screenshot shows a sample GUI for reference purpose

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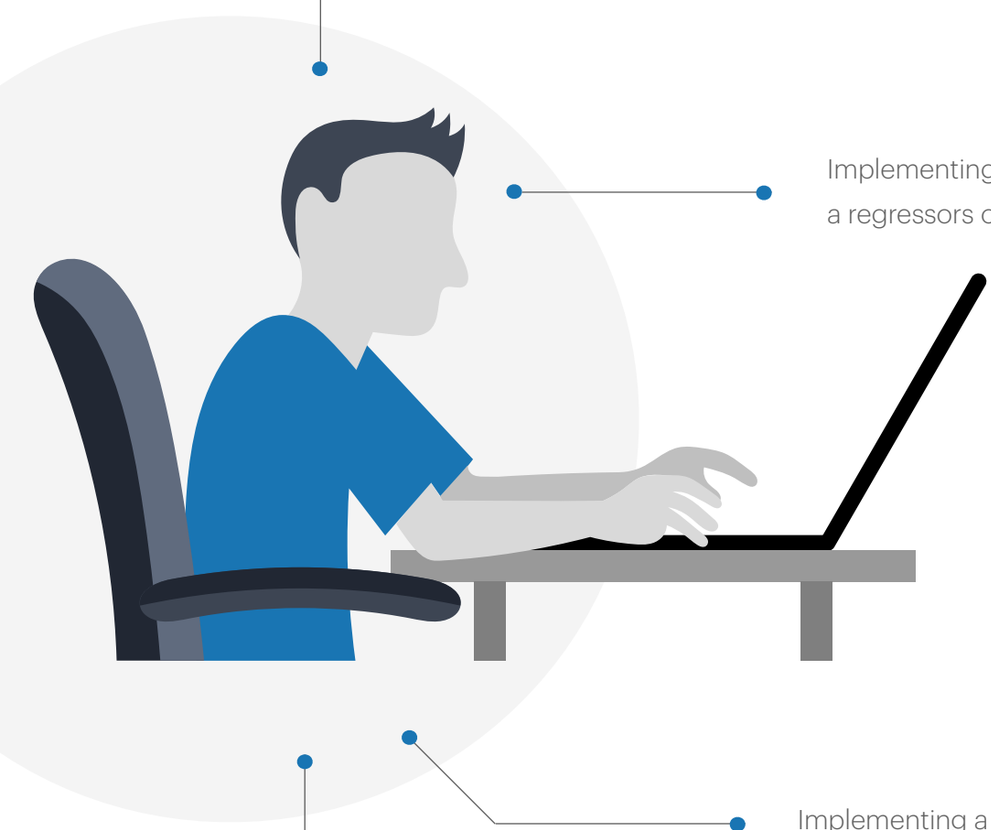
LEARNING OUTCOME

Hands on designing, training and testing a handmade a deep learning neural network.

Implementing a neural network as a classifier and a regressors on the same data.



Implementing a working GUI

Hands on storing the best model and its weights for future usage avoiding the need to retraining the model for future.

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" Put yourself in the shoes of an actual "

DATA SCIENTIST

THAT's YOU

Assume that you are working at the company which has received the above problem statement from internal/external client. Finding the best solution for the problem statement will enhance the business/ operations for your organisation/project. You are responsible for the complete delivery. Put your best analytical thinking hat to squeeze the raw data into relevant insights and later into an AIML working model.

PLEASE NOTE

Designing a data driven decision product typically traces the following process:

1. Data and insights:
   Warehouse the relevant data. Clean and validate the data as per the the functional requirements of the problem statement. Capture and validate all possible insights from the data as per the the functional requirements of the problem statement. Please remember there will be numerous ways to achieve this. Sticking to relevance is of utmost importance. Pre-process the data which can be used for relevant AIML model.
2. AIML training:
   Use the data to train and test a relevant AIML model. Tune the model to achieve the best possible learnings out of the data. This is an iterative process where your knowledge on the above data can help to debug and improvise. Different AIML models react differently and perform depending on quality of the data. Baseline your best performing model and store the learnings for future usage.
3. AIML end product:
   Design a trigger or user interface for the business to use the designed AIML model for future usage. Maintain, support and keep the model/ product updated by continuous improvement/training. These are generally triggered by time, business or change in data.

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IMPORTANT POINTERS

Project should be submitted as a single ".html" and ".ipynb" file. Follow the below best practices where your submission should be:

- ".html" and ".ipynb" files should be an exact match.
- Pre-run codes with all outputs intact.
- Error free & machine independent i.e. run on any machine without adding any extra code.
- Well commented for clarity on code designed, assumptions made, approach taken, insights found and results obtained.

Project should be submitted on or before the deadline given by the program office.

Project submission should be an original work from you as a learner. If any percentage of plagiarism found in the submission, the project will not be evaluated and no score will be given.

HAPPY LEARNING

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